

1. MEC 119.3 Applied Mechanics I (3-2-0)

Evaluation:

	Theory	Practical	Total
Sessional	50	-	50
Final	50	-	50
Total	100	-	100

Course Objectives:

This course has been developed to provide the basic knowledge of engineering mechanics to the students of engineering so that it would be beneficial to understand structural engineering. The knowledge of mechanics can utilized in wide range of engineering applications using Newton's laws of motion and mechanical equilibrium of different force system. This course shall be considered as a basic for all branches of Engineering of Pokhara University in first year of undergraduate program.

Course Contents:

1. **Introduction** (2 hrs)
 - 1.1 Definition and scope of Applied Mechanics
 - 1.2 Concept of Statics and Dynamics
 - 1.3 Concept of Particle
 - 1.4 Concept of Rigid, Deformed and Fluid Bodies
 - 1.5 Fundamental Concepts and Principles of Mechanics: Newtonian Mechanics
 - 1.6 System of Units
2. **Review of Coordinate System** (2 hrs)
 - 2.1 Cartesian Coordinate System
 - 2.2 Polar Coordinate System
 - 2.3 Cylindrical Coordinate System
 - 2.4 Spherical Coordinate System
 - 2.5 Review of Vector Algebra
3. **Forces acting on particles and rigid body** (7 hrs)
 - 3.1 Types of Forces: Point Force, Transitional and Rotational Force- Relevant Examples
 - 3.2 Resolution and Composition of Forces- Relevant Examples
 - 3.3 Principle of Transmissibility and Equivalent Forces- Relevant Examples
 - 3.4 Moments: Moment of a Force about a point and an axis- Relevant Examples
 - 3.5 Theory of Couples:: Relevant Examples
 - 3.6 Resolution of a Force into Forces and a Couple- Relevant Examples
 - 3.7 Resultant of Force and Moment for a System of Force: Examples
4. **Basic Concept of Static Equilibrium** (2 hrs)
 - 4.1 Concept of Load types, Load Estimation and Support Idealizations- Examples and Standard Symbols
 - 4.2 Free Body Diagram- Relevant Examples
 - 4.3 Physical Meaning of Equilibrium and its essence in structural application
 - 4.4 Equation of Equilibrium in Two/Three Dimensions
5. **Friction Forces** (3 hrs)
 - 5.1 Introduction
 - 5.2 Types of Friction and its Coefficients: Static and Dynamic
 - 5.3 Laws of Friction
 - 5.4 Angle of Friction

- 5.5 Engineering Examples of usage of Friction
6. **Center of Gravity, Centroid and Moment of Inertia** (6 hrs)
- 6.1 Concept and Calculation of Centre of Gravity and Centroid of Line/Area /Volume – Examples
- 6.2 Concept and Calculation of Second Moment of Area/ Moment of Inertia and Radius of Gyration- Relevant Examples
- 6.3 Use of Parallel Axis Theorem: Relevant Examples
7. **Analysis of Beam** (5hrs)
- 7.1 Introduction Beam
- 7.2 Use of statically determinant beam
- 7.3 Relationship between Load, Shearing Force and Bending Moment
- 7.4 Calculation and drawing of Axial Force, Shear Force and Bending Moment
8. **Analysis of Truss** (4 hrs.)
- 8.1 Introduction to determinant truss
- 8.2 Use of truss in engineering application
- 8.3 Analysis of force by method of joint
- 8.4 Analysis of force by method of section
9. **Kinematics of Particles and Rigid body** (6 hrs)
- 9.1 Rectilinear Kinematics: Continuous Motion
- 9.2 Position, Velocity and Acceleration of a Particle and Rigid body
- 9.3 Determination of Motion of Particle and Rigid body
- 9.4 Uniform Rectilinear Motion of a Particles
- 9.5 Uniformly Accelerated Rectilinear Motions of Particles
- 9.6 Curvilinear Motion of a Particle
- 9.7 Rectangular Components of velocity and Acceleration
- 9.8 Introduction of Tangential and Normal Components
- 9.9 Radial and Transverse Components
10. **Kinetics of Particles and Rigid body: Force and Acceleration** (6 hrs)
- 10.1 Newton's Second Law of Motion and Momentum
- 10.2 Equation of Motion and Dynamic Equilibrium: Relevant Examples
- 10.3 Angular Momentum: Rate of Change and Conservation
- 10.4 Motion of Various Particles and Relative Velocity
- 10.5 Equation of Motion- Rectilinear and Curvilinear
- 10.6 Rectangular Components: Tangential and Normal
- 10.7 Polar Coordinates: Radial and Transverse Components
11. **Moment and Energy in Rigid body** (2 hrs)
- 11.1 Introduction to Moment and Energy
- 11.2 Conservation of Linear and Angular Momentum

Text Books:

1. "Engineering Mechanics-Statics and Dynamics", Shames, I. H. 3rd ed., New Delhi, Prentice Hall of India, 1990.
2. "Mechanics of Engineers-Statics and Dynamics", F. P. Beer and E. R. Johnston, Jr. 4th Edition, Mc Graw-Hill Book Co., New York, USA (Asia Editions), 1987.

References:

1. "Engineering Mechanics-Statics and Dynamics", R.C. Hibbeler, Ashok Gupta, 11th edition. New Delhi, Pearson, 2009.
2. "Engineering Mechanics- Statics and Dynamics", I.C. Jong and B.G. Rogers.
3. "Engineering Mechanics- Statics and Dynamics", D.K. Anand and P.F. Connif.
4. "Engineering Mechanics of Solids", Egor .P. Popov, 2nd Edition, New Delhi, Prentice Hall of India, 1996.

5. "Engineering Mechanics- Statics & Dynamics", Dr. D.S. Kumar, S.K. Kataria & Sons, New Delhi, Reprint 2011.
6. Practice guide in Applied Mechanics, D. B. Pandit, Ramesh Khanal

6. MTH 121.3 Engineering Mathematics II (3 – 2 – 0)

Evaluation:

	Theory	Practical	Total
Sessional	50	-	50
Final	50	-	50
Total	100	-	100

Course Objective:

The main objective of this course is to provide the basic knowledge of three dimensional geometry, Calculus of several variables, differential equation, Laplace transform. After the completion of this course, students can use their knowledge in their professional course.

Chapter	Content	Hrs
1	Three Dimensional geometry : i. Review of direction cosines, direction ratios, Planes ii. Straight lines iii. Sphere and its tangent plane iv. Cone and cylinder(definitions, standard equation only)	12
2	Partial derivatives and Extreme values for function of two or more variables: i. Definitions, total derivatives, Chain rule, Eulers theorem for function of two or three variables, its application ii. Extreme values for two or more variables	6
3	Laplace transformation: i. Definition ii. Derivation of formulae iii. Application of laplace transform, iv. Inverse laplace transform v. Convolution theorem on laplace transform and application	8
4	Differential equation: i. Order and degree of differential equation ii. First order differential equation with their solutions (separable, reducible to separable form exact ness condition), linear and Bernoulies equation) iii. Second order differential equation (Homogeneous and non homogeneous) with constant coefficient as well as variable coefficients. iv. Initial value problem. v. Power Series solution vi. Legendres and Bessel equation with their solution, properties and application	13
5.	Double Integral: i. Definitions, Fubinis theorems (statement only) ii. Change of order, iii. Change Cartesian integral to equivalent polar integral iv. Area and volume by double integral	6

Text Books:

1. Engineering Mathematics II: Prof. D.D Sharma (Regmi), Toya Narayan Paudel, Hari Prasad Adhikari, Sukunda publication, Bhotahity, Kathmandu.
2. Advance Engineering Mathematics : Erwin Kreyszig.

Reference Books:

1. Calculus with analytical geometry: E.W. Swokowski.
2. Algebra: G.D Pant
3. Three Dimensional Geometry: Y.R Sthapit, B.C Bajracharya
4. Calculus and analytical geometry: George B Thomas, Ross L. Finney

7. MEC 109.2 Engineering Drawing (0-0-6)

Evaluation:

	Theory	Practical	Total
Sessional	-	50	50
Final	-	50	50
Total	-	100	100

Course Objectives:

1. To develop sketching, lettering and drafting skills
2. To draw projections, drawings of various geometric figures.
3. To draw assembly of machine parts.
4. To develop ability of preparing working drawings

Course Contents:

1. Instrumental Drawing, Practices and Techniques (12 hrs)

Equipment and metals, Description of drawing instruments, auxiliary equipment and drawing materials, Techniques of instrument drawing, pencil sharpening, securing paper, proper use of T-squares, triangles, scales, dividers, compasses, erasing shields, French curves, inking pens.

Freehand Technical Lettering

Lettering strokes, letter proportions, use of pencils and pens, uniformity and appearance of letters, freehand techniques, inclined and vertical letters and numerals, upper and lower cases, standard English lettering forms.

Dimensioning

Fundamentals and Techniques: size and location dimensioning, IS conversion; Use of scales, measurement units, reducing and enlarging drawings; General dimensioning practices: placement of dimensions aligned and unidirectional recommended practice, some 50 items.

2. Applied Geometry (24 hrs)

Plane geometrical construction: Bisecting and trisecting lines and angles, proportional division of lines, construction of angles, triangles, squares, polygons, constructions using tangents and circular arches. Methods of drawing standard curves such as ellipse, parabolas, hyperbolas, involutes, spirals, cycloid and helices (cylindrical and helical); Solid geometrical construction: Classification and pictorial representation of solid regular objects such as: prisms, square, cubical, triangular and oblique, Cylinders: right and oblique, Cones: right and oblique, Pyramids: square, triangular, oblique, truncated; Doubly-curved and warped surfaces: Sphere, torus, oblate ellipsoid, conoid, serpentine, paraboloid, hyperboloid

Basic Descriptive Geometry

Introduction: Application of descriptive geometry principles to the solution of problems involving positioning of objects in three-dimensional space; The projection of points, and planes in space; Parallel lines; True length of lines: horizontal, inclined and oblique lines; Perpendicular lines; Bearing of a line; Point view of end view of a line; Shortest distance from a point to a line; Principal lines of a plane; Edge view of a plane; True shape of an oblique plane;

Intersection of a line and plane; Angle between a line and a plane; Angle between two non-intersecting (skew) lines; Dihedral angle between two planes; Shortest distance between two skew lines.

3. **Theory of Projection Drawing** (24hrs)
Perspective projection drawing; Orthographic projection; Axonometric projection; Oblique projection; First and third angle projection;
Multi-view Drawings
Principal views: Methods for obtaining orthographic views: Projection of lines, angles and plane surfaces, analysis in three views; Projection of curved lines and surfaces; Object orientation and selection of views for best representation; Full and hidden lines. Orthographic drawings: Making an orthographic drawing, Visualizing objects from the given views; Interpolation of adjacent areas; True-length lines; Representation of holes; conventional practices.
- Sectional views**
Full section view; Half section; Broken section; Revolved section; Removed (detail) sections; Phantom of hidden section; Auxiliary sectional views; Specifying cutting planes for sections; conventions for hidden lines, holes, ribs, spokes.
- Auxiliary Views**
Basic concept and use of auxiliary views; Drawing methods and types of auxiliary views; Symmetrical and unilateral auxiliary views; Projection of curved lines and boundaries; Line of intersection between two planes; True size of dihedral angles; True size and shape of plane surfaces.
4. **Development and Intersections** (15hrs)
Development: General concepts and practical considerations, Development of a right or oblique prism, cylinder, pyramid and cone; Development of truncated pyramid and cone; Triangulation method for approximately developed surfaces; Transition pieces for connecting different shapes; Development of a sphere; Intersections: Lines of intersection of geometric surfaces; Piercing point of a line and a geometric solid; intersection lines of two planes; Intersection of prisms and pyramids; Intersection of a cylinder and an oblique plane; Intersection of a sphere and an oblique plane; Constructing a development using auxiliary views; Intersection of two cylinders; Intersection of a cylinder and cone.
5. **Machine Drawing** (15hrs)
Introduction: production of complete design and assembly drawings; Fundamental techniques: size and location dimensioning; placement of dimension lines and general procedures; standard dimensioning practice (IS system); Limit dimensioning: nominal and basic size, allowance, tolerance, limits of size, clearance fit, interference fit; basic hole system and shaft systems; Thread and standard machine assembly elements: screw threads: ISO standards, representation and dimensioning; Fasteners: type and drawing representation, keys, collars, joints, springs bearings; Assembly drawings: drawing layout, bill of materials, drawing layout, bill of materials, drawing numbers.
- Laboratory Work:**
Freehand technical lettering and use of drawing instruments; Dimensioning; Geometrical and Projection drawing; Descriptive geometry; Projection and multiview drawings; Sectional views; Auxiliary views, Freehand sketching and visualization; Development and intersections; machine and assembly drawings.
- Reference Books:**
8. Luzadder, *Fundamentals of Engineering Drawing*, Prentice Hall of India Ltd., 8th edition, 1981.
 9. French, C.J. Vierck and R.J. Foster, *Engineering Drawing and Graphic Technology*, McGraw-Hill, 1981.
 10. Machine drawing P.S. Gill, S.K. Kataria and Sons, India, 7th Edition, 2008.

10. CMP 104.3 Object Oriented Programming in C++ (3-1-3)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

- To familiarize with Object Oriented Concept.
- To introduce the fundamentals of C++
- To enable the students to solve the problems in Object Oriented technique
- To cope with features of Object Oriented Programming

Course Contents:

Chapter	Content	Hrs.
1	Thinking Object Oriented Object oriented programming a new paradigm, a way of viewing world agent, types of classes, computation as simulation, coping with complexity, nonlinear behavior of complexity, abstraction mechanism	4
2	Classes and Methods: Review of structures, classes and inheritance, state, behavior, method, responsibility, encapsulation, data hiding, Functions: friend function, inline function, static function, reference variable, default argument	7
3	Message, Instance and Initialization Message, message passing formalization, message passing syntax in C++, mechanism for creation and initialization (constructor and its types), Issues in creation and initialization: memory map, memory allocation methods and memory recovery	6
4	Object Inheritance and Reusability Introduction to inheritance, Subclass, Subtype, Principle of Substitutability; Forms of polymorphism and their implementation in C++, inheritance merits and demerits, composition and its implementation in c++, The <i>is-a</i> rule and <i>has-a</i> rule, Composition and Inheritance contrasted, Software reusability	9
5	Polymorphism Polymorphism in programming language, Varieties of polymorphism, compile time polymorphism, function overloading, operator overloading, type conversion, polymorphic variable, run time polymorphism, object pointer, this pointer, virtual function, overriding, deferred method, pure polymorphism.	8
6	Template and generic programming Generic and template functions and classes, cases study: container class and the	4

standard template library, Exception handling

7 Object oriented Design

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Reusability implies non- interference, Programming in small and programming in large, components and behaviors, role of behaviors in OOP, CRC, sequence diagram, Software components, formalizing the interface, interface and implementation, Design and representation of components, coming up with names, implementation components, integration of components

Laboratory Work

There shall be 20 exercises in minimum, as decided by the faculty. The exercises shall encompass a broad spectrum of real-life and scientific problems, development of small program to the development of fairly complex subroutines, programs for engineering applications and problem solving situations. Laboratory assignments will be offered in groups of two to four for evaluation purpose. In general, the Laboratory Work must cover assignments and exercises from the following areas:

1. Data types – control structures, functions and scoping rules.
2. Composite data types, C++ strings, use of " Constant " keyword, pointers and references
3. Classes and data abstraction
4. Inheritance, abstract classes and multiple inheritance
5. Friend functions, friend classes and operator overloading.
6. Static class members
7. Polymorphism, early binding and late binding
8. C++ type conversion
9. Exception handling
10. Function templates, class templates and container classes.

Textbooks:

1. Budd, T., *An Introduction to Object Oriented Programming*, Second Edition, Addison-Wesley, Pearson Education Asia, ISBN: 81-7808-228-4.
2. R. Lafore, *Object Oriented Programming in Turbo C++*, Galgotia Publications Ltd. India, 1999

Reference Books:

1. E Balaguruswamy, *Object Oriented Programming with C++*, Third Edition
2. Tata McGraw-Hill ISBN:0-07-059362-0, Parson David, *Object Oriented Programming with C++*, BPB Publication\ISBN817029-447-9

11. PHY 102.4 Physics (4-2-2)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

The main objectives of this course are:

1. To apply the theory of simple Harmonic motion in different elastic systems.
2. To apply theory of wave propagation and knowledge of resonance.
3. To apply and analyze the Optical properties in different optical systems.
4. To make use of fundamentals of electromagnetic equipment.
5. To use the knowledge of basic physics in different engineering fields.

Chapter	Content	Hrs
1	Mechanical Oscillation Introduction and equation of Simple Harmonic Motion, energy in Simple Harmonic Motion, oscillation of mass –spring system, compound pendulum	4
2	Wave motion Introduction of wave, wave velocity and particle velocity, types of waves, equation, energy, power and intensity of plane progressive wave, standing wave and resonance.	4
3	Acoustics Reverberation of sound, absorption coefficient, Sabines formula, introduction, production and applications of ultrasonic wave	4
4	Physical Optics Interference: introduction, coherent sources, interference in thin films due to reflected and transmitted light, Newton's Ring (3) Diffraction: introduction, fraunhofer diffraction at single slit and double slit diffraction grating (2) Polarization: introduction, double refraction, Nicol prism, optical activity, specific rotation, wave plates (3)	8
5	Laser and Fiber Optics Introduction of laser, spontaneous and stimulated emission, optical pumping, He-Ne laser, Ruby Laser, use of laser, Propagation of light waves, types of optical fiber, applications of optical fiber	4
6	Electrostatics Electric charge, electric force, electric flux, electric potential, Gauss law and its applications, electric field intensity and electric potential due to dipole, electric potential due to quadrupole, capacitors, electrostatic potential energy, dielectrics and gauss law charging and discharging of capacitor	8

7	Electricity and magnetism	10
	Electric current, resistance, resistivity and conductivity, atomic view of resistivity, magnetic field, magnetic force, Lorentz force, Hall effect, Biot-Savart's law and its applications, force between two parallel conductors, Ampere's circuital law and its applications, Faraday's law of electromagnetic induction, self-induction R-L circuit, energy stored in magnetic field and magnetic energy density	
8	Electromagnetism	9
	LC oscillation, Damped oscillation, forced oscillation and resonance, Maxwell's equations displacement current, wave equations in free space, continuity equation, E and B fields, Poynting vector, radiation pressure	
9	Photon and matter waves	4
	Photon, group velocity and phase velocity, De Broglie wavelength, Schrodinger wave equation, one dimensional potential well, tunneling effect	
10	Semiconductors and super conductivity	5
	Introduction, types of semiconductors Doping, P-N Junction, Metal- semiconductor junction, junction breakdown, junction capacitance, electrical conduction in metals, insulators and semiconductors according to band theory of solids, introduction to superconductor	

Textbooks:

3. Fundamental of Physic by Robert Resnick and David Haliday
4. A Text Book of Engineering Physics, T. R. Lamichane
5. A text book of optics by Subramanyam and Brijlal
6. Modern physics by R. Murugason

Reference Books:

16. Concept of physic by H.C Verma
17. Modern Engineering Physic by A.S Basudeva
18. Electronics by B.L Thereja
19. Principles of Electronics, V. K. Meheta

Laboratories:

1. To determine the acceleration due to gravity & radius of gyration by single bar pendulum.
2. To determine the frequency AC mains by using son meter apparatus
3. To determine the wave length by using diameter of Newton's ring
4. To determine the wave length of laser light by using diffraction grating
5. To determine the value of Modulus of Elasticity of the material given and Moment of Inertia of Circular disc using torsional pendulum
6. To determine the capacitance of given capacitor by charging and discharging through resistor
7. To determine the low resistance of a given wire and resistance per unit length of the wire by using Carey-foster bridge
8. To plot a graph current and frequency in an LRC series circuit and to find: i) the resonance frequency
ii) the quality factor

Lab textbook: B. Sc Practical Physics by C. L. Arora

12. MEC 189.2 Thermal Science (2-1-2)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

The broad objective of this course is to provide working knowledge of theories and applications of thermal science.

The specific objectives of the course are:

1. To make able to apply laws of thermodynamics in various systems.
2. To make able to distinguish the cycles in various engines, and pumps.
3. To make able to calculate energy/quantity of heat transfer by conduction and radiation.

Chapter	Content	Hrs.
1	Concept and definitions Thermodynamics, Applications of thermodynamics, Thermodynamic system, Macroscopic and microscopic and microscopic Approaches, Properties and state of a substance: Thermodynamic properties and types, State, path, process, cycle. • Processes (definition, characteristics, features, Examples): Quasi-equilibrium(states) process, Reversible process, Irreversible process, • Specific volume, Pressure, Atmospheric pressure, Gauge pressure, Absolute pressure • Equality of temperature: Zeroth law of thermodynamics	2
2	Properties of pure substances • Pure substance: Homogenous in composition, Homogenous in chemical aggregation, Invariable in chemical aggregation • Vapour-liquid solid phase equilibrium in pure substance: Steam generation(formation) process from ice to steam, Wet steam and quality, T-v diagram of water, P-v Diagram of water, P-t diagram of water • Equations of state for a simple compressible substance: • Tables and diagrams of thermodynamic properties • Determination of Specific volume, Specific enthalpy and Specific entropy of wet and superheated steam	3
3	Work and heat Definition of work: in mechanics and in thermodynamics, Work done in quasi-equilibrium process Displacement work, Work done in different reversible processes: • Isochoric process • Isobaric process • Isothermal process • Polytrophic process • Definition of heat: comparison between heat and work,	2

4 First law of thermodynamics	4
- First law for cycle: First law for closed system undergoing a cycle, Verification of this law by wheel paddle experiment - First law for process: Difference between stored and internal energy, Stored energy - Internal energy: Joule's law and its verification - Enthalpy - Specific heats: Specific heat capacity of gas at constant volume, Specific heat capacity of gas constant pressure - First law as a rate equation: - Conservation of mass and the control volume - First law for control volume Steady state steady flow process: Assumption, Steady state steady flow energy equation (SFEE), Application of SFEE: Heat exchanger, nozzle, diffuser, turbine, Rotary compressor, Throttling device, Boiler - Uniform state uniform flow process: Assumptions, 1st law for uniform state uniform flow process	
5 Second law of thermodynamics	3
- Heat engines: 4 components diagram and the schematic diagram, efficiency - Refrigerator and heat pump: 4 components diagram and schematic diagram, COP of refrigerator and heat pump Second law: Limitation of first law of thermodynamics, Kelvin-Planck statement, Clausius statement - Equivalence of Kelvin-Planck and Clausius statements: - Factors causing irreversibility - Carnot theorem - Thermodynamic temperature scale	
6 Entropy	4
Inequality of Clausius, Entropy as a property of a system, Entropy of pure substance, Entropy change in reversible process, lost work, principle of increase of entropy, Entropy change of an ideal gas, the poly-tropic process for an ideal gas, concepts of reversibility, irreversibility and availability	
7 Some Power Cycles	4
- Vapor Power Cycles: Rankine cycle (working process, efficiency, Effect of pressure and temperature on Rankine cycle) - Air Standard Cycles: Air standard cycles: Carnot cycle (Working processes & Efficiency), Brayton cycle (Working processes & Efficiency) - Internal combustion engines: Otto cycle (Working processes & Efficiency), Diesel cycle (Working processes & Efficiency), Comparison between Otto and diesel cycle	

8 Heat transfer

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- Modes of heat transfer: Conduction, Convection, radiation
- Conduction : Fourier's law (Statement, Mathematical modeling, Assumption for this laws, Thermal conductivity
- One dimensional steady state heat conduction: Through a plane, Through a hollow cylinder, Through a hollow sphere
- Composite wall: Heat flow through multilayer plane slabs, Numerical on wall of planes, cylinders and spheres in series.
- Thermal resistance and conductance: Electrical analogy of the conduction heat flow
- Overall heat transfer co-efficient: Heat transfer through a plane slab separating two fluid media
- Basic laws of radiation: Emissive power and emissivity, Stefan-Boltzmann's law, Kirchoff's law, Wei's displacement law
- Black and gray bodies: Reflectivity, absorptive and transmissibility, Black and grey bodies
- Radiant exchange between infinity parallel planes
- Newton's law
- Mechanism of forced and free convection
- Dimensionless parameters: Reynold's number, Nusselt's number, Prandtl's number

9 Introduction to Refrigeration System

1

Introduction, Refrigeration cycle

Laboratory Work:

1. To measure the pressure, specific volume and temperature
2. To find out the efficiency of a compressor.
3. To measure the rate of heat transfer by conduction.
4. To measure performance of a small internal combustion engine
5. To measure the heat transfer by thermal radiation.
6. To measure the performance of a Refrigeration/Heat pump

Textbooks:

1. Howell J.R. and R.O. Buckius, *Fundamentals of Engineering Thermodynamics*, McGraw-Hill Publishers, 1994.

Reference Books:

1. Van Wylen, G.J. and Richard E. Sonntag, *Fundamentals of Classical Thermodynamics*, Wiley Eastern Limited, New Delhi, 1989.
2. Bayazitoglu, Y. and M. Necati Ozisik, *Elements of Heat Transfer*, McGraw-Hill Book Company, 1998.
3. Kreith, F., *Principles of Heat Transfer*, International Text book Company, Scranton Pennsylvania, 2nd Edition, 1965.